Customer Shopping Behaviour Analysis

Business Problem

A leading retail company wants to better understand its customer's shopping behaviour in order to improve sales, customer satisfaction and long term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer and repeat purchases.

You are tasked with analyzing the company's consumer behaviour dataset to answer the following overarching business question.

“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketingg and product strategies?”

1. Project Overview
The project analyzes customer shopping transactional data of 3,000 purchases of different product category. The main goal of the analysis is to uncover insights into spending pattern, customer segments, product purchases and subscription behaviour inorder to guide strategic business decisons.

2. Dataset Summary
   - Rows:
   - Colmns:
   - Null_values: 37 missing values in review rating column

3. Exploratory Data Analysis using Python
I began with data preprocessing and cleaning using pandas
Initial Exploration: Used df.info() to check structure of the data and df.describe() for both numerical and categorical variables summary statistics

Getting sumary statistics of both numerical and categorical columns

```
df.describe(include ="all")
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Identifying missing values: A total of 37 values were null in the review rating column and the missing values were handled by identifying the median value for each product category.

```
df.isnull().sum()
```

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

Replacing the missing values in Review Rating with the median review rating for each category

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
```

- Column Standardization: Renamed columns to snake case for better readability and documentation.

- Feature Engineering:
    - Created age_group column by binning customer age

- Created a purchase frequency days column from purchases data to get the number of days

- Data Consistency Check: Verified if discount_applied and promo_code_used were redundant, and dropped_promo_code used column.
- Data Base Integration: Connected python script to MYSQL, and loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis Using MYSQL.

Major business questions was answered, using SQL queries

1. Revenue by gender: compared total revenue generated by male vs female customers.

Result Grid | Filter Rows

gender	total_revenue
Male	157890
Female	75191

2. Identified high spending discount users- identified customers who used discounts but still spent above the average purchase amount.

Result Grid | Filter Rows:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90

3. Identified top 5 products by rating: discovered products with the highest average review ratings.

Result Grid | Filter Rows: | Export:

category	item_purchased	average_product_rating
Accessories	Gloves	3.86
Footwear	Sandals	3.84
Footwear	Boots	3.82
Accessories	Hat	3.8
Accessories	Handbag	3.78

4. Shipping type comparison: compared average purchase amounts between standard and express shipping

Result Grid | Filter Rows:

shipping_type	average_purchase_amount
Express	60.48
Standard	58.46

5. Subscribers vs non subscribers - Compared average spend and total revenue across subscription status.

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_status	total_customers	average_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

6. Discount-Dependent Products - Identified 5 products with the highest percentage of discounted purchases.

Result Grid | Filter Rows:

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation: Grouped customers into New, Returning and Loyal segments based on purchase history.

Result Grid | Filter Rows:

customer_segment	Number_of_Customers
Loyal	3116
Returning	701
New	83

8. Top Products per Category: Listed the most purchased products within each category.

Result Grid | Filter Rows: | Export:

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

9. Repeat Buyers & Subscriptions : Checked if customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

10. Revenue by Age Group: Calculated total revenue contribution of each age.

age	total_revenue
49	5552
69	5484
25	5372
41	5282
54	5282
57	5200
28	5104
19	4941
50	4930

5. Dashboard: An interactive dashboard was built in PowerBI to convey insights visually

6. Business Recommendation.

- Leverage high demand for products in top performing category for saes and revenue like clothing to cross sell secondary products.
- Focus growth efforts on customer retention and repeat purchases, through targeted lifecycle marketing.
- Implement strategic marketing tailored to female demographics

- Use most purchased products in each product category as primary marketing hooks, feature these specific items on home pages, ad creatives and search ads
- Reward loyal customers with exclusive access and referral incentives.
- Convert new and returning customers into loyal buyers using targeted post purchase email sequences, personalized cross-sells and progressive rewards.
- Focus efforts on high revenue age groups and express-shipping users.